

SIMATS ENGINEERING

ASSESSMENT TOOL 1-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING

COURSE CODE: ITA04

COURSE OUTCOME COVERED

CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.

(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

Code of Conduct:

I Rahima Banu.N - 192525009 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rahima Banu.N

PSEUDOCODE WRITING TASK

Q: Variables and Data types Analysis:

Pseudocode:

START

Set num = 20

Set name = "Rahima"

Set status = TRUE

Sum = num + 10

result = status AND FALSE

Display num

Display name

Display status

Display sum

Display result

STOP

OUTPUT:

20

Rahima

TRUE

30

FALSE.

Q2: Data Structures Implementation

PSEUDOCODE:

START

Create Vector = [10, 20, 30]

Create List = ["Rahima", 19, TRUE]

Create Matrix = [[1, 2], [3, 4]]

Display Vector[2]

Update Vector[2] = 25

Find sum of Vector

Display Vector

Display List

Display Matrix

STOP

OUTPUT:

Vector: 10 25 30

List:

Rahima

19

TRUE

Matrix:

1 2

3 4

Sum = 65

mix and Away Operations:

PSEUDOCODE:

START

Create Matrix A

Create Matrix B

Add A and B

Multiply A and B.

Create 3D array

Display addition

Display Multiplication

Display array

STOP.

OUTPUT:

Matrix Addition:

6	8
10	12

Matrix Multiplication:

19	22
43	50

3D array Created Successfully.

Q4: Data Frame Construction And Manipulation:

PSEUDOCODE:

START

Create Data Frame (Name, Marks)

Add Grade Column

Delete Grade Column

Modify Marks

Display Data Frame

Display Summary

STOP

OUTPUT:

Name	Marks
Rahma	95

Q5: Functions and Logical Operations

PSEUDOCODE:

START

Create Function Square(x)

Return $x \times x$

Call Square(5)

Create Logical Vector

Perform Logical AND

Display Result

STOP

OUTPUT:

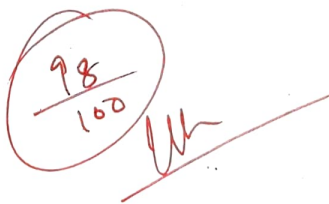
Square = 25

Logical Vector

TRUE
FALSE
TRUE

Good work 11/7/26

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately ✓	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution ✓	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules ✓	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow ✓	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach ✓	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient



 98 / 100